PAWN

From expendable peon to battlefield legend

First-person · Asymmetric RTS hybrid · Worker-sim · Cartoon fantasy · 3D · Unity · Single & Multiplayer

Pawn is a first-person, physics-driven take on the classic real-time strategy formula. You are **not** the commander. You are one lowly worker — a peon — standing in the middle of a full-scale war that an **AI "General"** runs all around you. Mine gold from inside the mine, chop and carry timber in your arms, build, and — only as a last resort — fight. Obey the General's orders, or defy him and risk his fury. Earn XP, grow a talent tree, and gently **influence** the General's strategy to turn the tide. Die, and you respawn as a fresh recruit with your in-match talents reset — but your **permanent progression** survives every battle.

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Version: 1.0 · **Status:** Draft for MVP scoping · **Working title:** PAWN (working title)
Target platforms: Windows (MVP) → Android & Steam Deck (post-MVP)
Engine: Unity (URP) · **MVP race:** Orcs (peon) · **Language:** EN/FR

◆ How to use this document

This document is written to be handed to an AI coding agent (or a human developer) as the single source of truth for the game. It separates **what** the game is (design) from **how** to build it (technical), and it clearly fences the **MVP** from later phases.

> **Conventions used throughout**
>
> - MVP = required for the first playable version. Deferred = later phase. Cut = out of current scope.
> - All numbers (capacities, damage, timings, costs) are **initial tuning values** — treat them as a starting point, kept in data/config, never hard-coded.
> - Diagrams marked MOCKUP are original UI/system illustrations to build from. Items marked REFERENCE are external examples for look & feel only.
> - The RTS layer (buildings, units, costs) is **inspired by the structure** of classic RTS games but uses original naming and slightly different stats.

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1 Overview & Vision

1.1 One-sentence pitch

Live a full real-time strategy battle from the inside: you are a single worker obeying (or defying) an AI General, mining, building and fighting your way from disposable peon to legend — with permanent growth that persists between matches.

GENRE	PERSPECTIVE	PLATFORMS	ENGINE
Asymmetric RTS / FPS hybrid	**3D, first-person**	**Win → Android / SteamDeck**	**Unity (URP)**

PLAYERS	MVP RACE
1 (solo) · up to 6 (PvP)	**Orcs (peon)**

1.2 Genre definition

Pawn is a **first-person asymmetric real-time-strategy hybrid**. Two games run simultaneously and are deeply coupled:

- **The RTS layer** (macro): an **AI "General"** plays a full classic RTS — base building, economy, tech tree, unit production, heroes, scouting, and combat. This layer looks and behaves like a traditional top-down strategy game, but the human player does **not** control it directly.
- **The first-person layer** (micro): the human player controls **one worker unit** in that world, in first person, with full body awareness, doing the physical labor of the war (mining, chopping, hauling, building, fighting).

The two layers meet in two places: the worker **executes** the General's orders, and the worker can **influence** the General's higher-level decisions. The result is a game about **agency within hierarchy**: you are one cog, but a cog with free will and a growing voice.

1.3 The fantasy (what it should feel like)

Stepping out of the starting camp feels like being dropped into the middle of someone else's match. Trees loom around you; a gold mine glows in the distance; the General's quest panel tells you to chop wood. You grab an axe, and a tree **physically falls** when you finish it — you feel the weight. You carry the logs in your arms back to the stockpile. Later you're sent to colonize a distant mine; on the way, enemies ambush you and you swing your pickaxe for your life. The General is making mistakes — leaving a rich mine uncolonized, forgetting to train warriors — and you nudge him. By endgame you are no longer just a peon; you are the reason your side won.

1.4 Design pillars

Pillar	Meaning	Implication
1. One cog, not the machine	You are a single worker inside a larger war you don't command.	The RTS runs with or without you; your contribution is real but partial.
2. Obey or defy	Orders can be followed or ignored — with consequences.	A loyalty system measures obedience; zero loyalty = the General turns on you.
3. Influence, don't command	You shape the General's decisions indirectly, like a pet nudging its owner.	A communication wheel + trust system; influence is never guaranteed.

4. Tactile labor	Gathering and building must feel physical and satisfying.	Physics-driven trees/mining; visible carrying; strong feedback.
5. Zero to legend	Progression from weak worker to powerhouse, within and across matches.	In-match talent trees + persistent meta-progression that survives death.

1.5 Target platforms & priorities

Platform	Priority	Notes
Windows (PC)	MVP	Primary target. Keyboard/mouse + gamepad.
Steam Deck	Post-MVP	Gamepad-first; verified against Steam Deck performance envelope.
Android	Post-MVP	Touch + gamepad; significant perf/UI work. Deferred deliberately.

1.6 Engine & tech stack

- **Engine:** Unity 2022 LTS (or newer), Universal Render Pipeline (URP).
- **Networking:** Unity Netcode for GameObjects (or equivalent) — P2P, see §15.
- **Physics:** Unity Physics + ragdolls; custom harvesting physics where needed (see §12).
- **Content:** Data-driven via ScriptableObjects / JSON (units, buildings, talents, meta upgrades).
- **Art style:** stylized / hand-painted cartoon (see §13).

1.7 Art style & tone

Stylized **cartoon fantasy** in the spirit of World of Warcraft: exaggerated proportions, chunky silhouettes, saturated colors, readable at a distance, with a light-but-not-comical tone. Serious stakes, playful visuals. Reference media is listed in §21.

1.8 Definition of “fun” (success criteria for the prototype)

- Chopping a tree and hauling the logs feels **good** (weight, sound, visible progress) — a player should want to keep doing it.
- Receiving an order and choosing to obey or ignore is a **meaningful** decision with legible consequences.
- Nudging the General into a better decision feels like **winning indirectly** — a satisfying “I told you so” moment.
- Dying feels **fair and motivating** (you keep your permanent progress), not punishing.

2 Game Modes

The game ships with three modes. The **core match flow is identical** across all three (the worker loop in §3); only the wrapper (story, progression, and who is on the other side) changes.

2.1 Campaign — “The Rise” (scripted)

A story-driven, single-player campaign. You play a single peon who rises from an anonymous grunt to a living legend — a “Solo Leveling”-style power fantasy, but **progressive and earned** rather than instant. The world reacts to your growing reputation: allies are shocked, enemies begin to fear and specifically hunt you, and the General increasingly relies on you.

- **Protagonist:** an Orc peon (MVP). Player controls only this worker.
- **Structure:** a sequence of missions/chapters, each a full RTS battle wrapped in a narrative framing (objective, intro/outro dialogue, set-piece moments).
- **Progression:** permanent (meta) progression applies and is the main long-term hook; in-match talents reset each mission as normal.
- **Narrative:** the General is a recurring character; the campaign explores the obedience/free-will and influence themes explicitly.
- Deferred Full voice acting / cinematic cutscenes — MVP uses dialogue text + simple camera staging.

2.2 Free Play — “Skirmish” (no scenario)

A non-scripted mode in the spirit of “pick a map and go”. No story; you select a map (and optional difficulty/loadout), play a standard match to completion, and earn meta progression to **unlock** permanent upgrades, equipment and new maps.

- **Purpose:** replayable loop for unlocking permanent content (the “Vampire Survivors” unlock loop, but the match itself is a normal RTS battle, not an endless horde).
- **Variety:** different maps (starting layout, mine placement, terrain), difficulty tiers that scale the enemy General’s skill and error rate.
- **Win:** destroy the enemy base (same as everywhere). MVP

2.3 Multiplayer — “War” (PvP)

Competitive 1v1 matches. Each side has its own AI General running the RTS; **human players are always workers** (peons) — there is **no human-controlled General**.

- **Players:** up to **3 humans per camp** (6 max on a 1v1 map), each controlling their own worker.
- **Teams:** workers on the same camp cooperate (shared objective); the opposing camp’s workers are enemies.
- **Netcode:** P2P (see §15).
- Deferred Ranked matchmaking, spectators, reconnect.

2.4 Mode comparison

	Campaign	Free Play	Multiplayer
Scenario / story	Yes (chapters)	No	No
Opponent General	AI	AI (scaled difficulty)	AI (one per side)
Human players	1 (solo)	1 (solo)	up to 3 / camp
Meta progression	Yes	Yes (main unlock loop)	Yes
MVP	MVP	MVP	Deferred (post-MVP)

MVP sequencing note

Campaign and Free Play share the same core systems, so they are built together in the MVP. Multiplayer is deliberately deferred because netcode and P2P are a large, independent effort (see §15, §17, §18).

3 Core Loop

3.1 The worker loop (minute-to-minute)

1. **Receive** an order from the General (quest panel + optional sound/minimap ping).
2. **Decide**: obey, ignore, or (later) try to influence instead.
3. **Act**: gather resources, construct/repair, haul goods, or fight — all first-person, physics-driven.
4. **Earn XP** from gathering, kills, or accepted influence → level up → spend a talent point.
5. **Influence** the General via the communication wheel to fix his mistakes or steer strategy.
6. **Survive**: manage health/stamina, avoid or win fights, watch loyalty & trust.

3.2 The three loops

Loop	Timescale	Reward
Micro (gather → haul → deposit)	~30 s – 2 min	Resources for the RTS; a trickle of XP; tactile satisfaction.
Match (level 1→30, talent build, influence the war)	~15–40 min	Win/lose; end-of-match meta XP; the fantasy of shaping a battle.
Meta (permanent upgrades between matches)	Many matches	Persistent power: better tools, stats, abilities, equipment.

MOCKUP

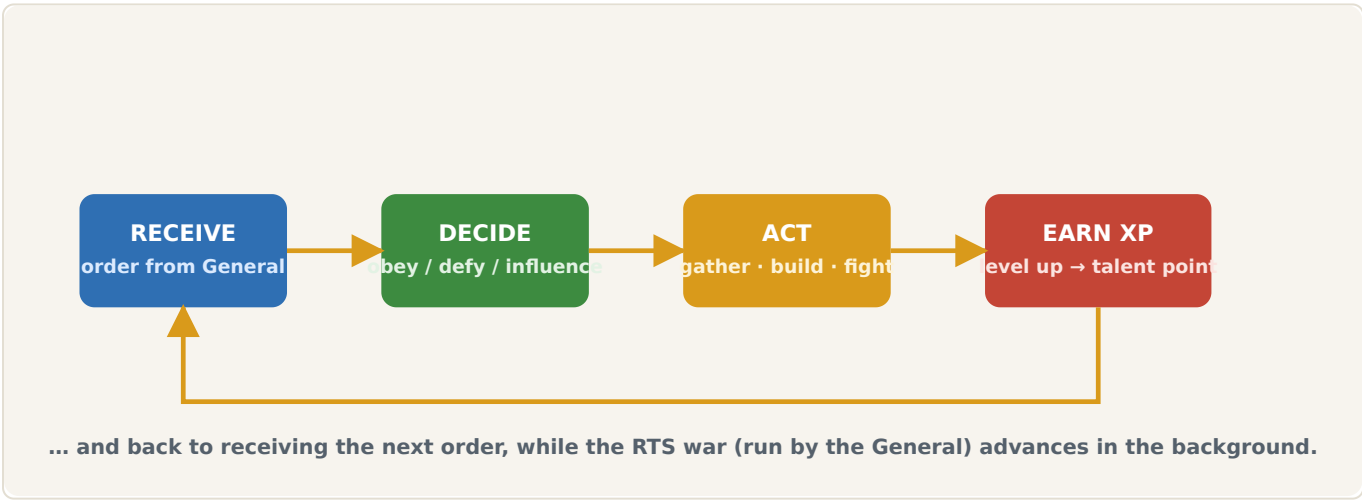


Fig. 3.1 — The core worker loop. The RTS war runs continuously in parallel, independent of the player's actions.

3.3 Session loop (one match)

- **Opening:** a few workers, a gold mine and nearby timber; the General begins expanding.
- **Mid-game:** tech-up, colonize distant mines, hero recruitment, first skirmishes; the player increasingly shapes the General's choices.
- **Endgame:** large armies clash; the player is a high-level, battle-hardened worker; one base falls.
- **End:** match ends when an enemy base is destroyed → meta XP awarded → back to mode select.

4 The Player — The Worker

4.1 Role & fantasy

The player is a single **worker unit** (MVP: an Orc **peon**). You are one of several workers the General starts with. You have a body, tools, and — crucially — **free will**. The game never takes control away from you; orders are requests you can honor or ignore.

4.2 First-person body awareness

Strong body awareness is a core requirement. In first person you must be able to see:

- **Hands & arms** — visible while walking, carrying, swinging tools, and in combat.
- **Feet & legs** — visible when looking down.
- **Torso** — visible at the edges of the view.
- **Carried items** — the **backpack** (gold) is visible on the back/shoulder; **logs** (wood) are visibly held in both arms, blocking part of the view when full.
- **Shadow & footsteps** — a real-time shadow and footstep feel ground the player in the world.

4.3 Movement & traversal

Action	Detail
Walk / run	Default locomotion; speed is reduced while carrying (see 4.4).
Sprint	Consumes stamina. Base skill, upgradeable via talents/meta.
Jump	Simple jump; jump height is a permanent meta upgrade.
Dodge / roll	Short dodge roll with brief invulnerability frames; unlocked via meta progression.
Climb	`Cut` — no climbing; traversal is horizontal + jump only.

4.4 Carrying & resources

Only **two** resources are gathered by the player; each uses a distinct, visible carrying method:

Resource	Carried in	Capacity (base)	Speed penalty
Gold	Backpack (on the back)	10 units	Small, scales with load
Wood	Both arms (logs held in front)	10 logs	Small, scales with load

- Carrying is **mutually exclusive** — you carry gold *or* wood at a time, not both.
- Deposit at the nearest **drop-off** (town hall or a storage building) to bank resources into the General's economy.
- Capacities are **upgradeable** via in-match talents and permanent meta upgrades.
- While carrying, tools are stowed (you cannot mine/chop/fight with full arms until you drop or deposit).

4.5 Health, stamina, death & respawn

- **Health (HP):** standard hit points; increased by talents and meta. Regenerates slowly out of combat.
- **Stamina:** spent by sprint, roll, and power attacks; regenerates when not spent.
- **Death:** when HP reaches 0 you die. On death, **all in-match talents and levels reset** — you respawn as a fresh worker.
- **Respawn:** after the active clash ends, the General trains a new worker (standard unit cost & a short timer). You keep your **permanent** meta progression.

- **Fate of carried goods:** dropped on the ground at death; can be recovered by you or another worker (or stolen by an enemy).

4.6 Combat (a last resort, but inevitable)

- **Base weapons:** the **axe** and **pickaxe** double as melee weapons (lower damage than dedicated weapons).
- **Weapon pickup:** you can pick up weapons dropped by fallen soldiers (melee or ranged) — a permanent meta unlock.
- **Disarm / steal:** with the right skill, you can **wrestle a weapon** (or later a **mount**) from an enemy or ally mid-fight.
- **Melee & ranged:** both are supported; the worker can use picked-up ranged weapons (bows/crossbows) with limited ammo.
- **Magic:** unlockable in two ways — (a) study at the **caster building** (the worker is unavailable for a duration while studying), or (b) via permanent meta unlocks / in-match skills.
- **Combat role:** the worker is **not** a front-line soldier; combat should feel desperate and risky, not dominant — at least until deep progression.

4.7 The HUD

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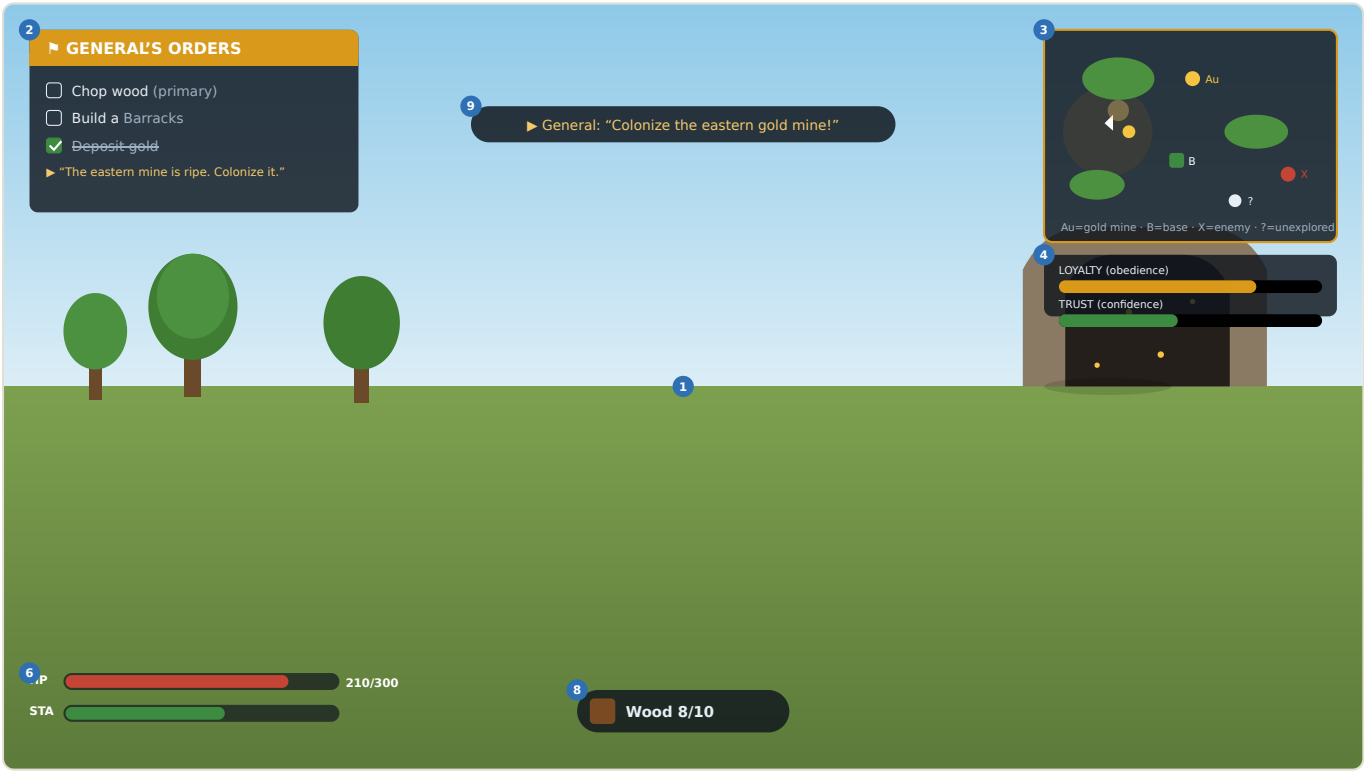


Fig. 4.1 — First-person HUD mockup. **1** crosshair · **2** General's order panel (quest list) · **3** minimap with gold mines (Au), base (B), enemy (X), player ping · **4** loyalty gauge (obedience) · **5** trust gauge (confidence) · **6** health · **7** stamina · **8** carried-resource indicator · **9** General's message toast.

5 Resources & Gathering

5.1 Gold — mined from inside

Gold is **not** picked up from the surface: you must **enter the mine** and strike ore veins with your pickaxe. Each hit yields gold, which fills your **backpack**. When full, you haul it back to the nearest drop-off. Colonizing distant mines (walking there, clearing it, and establishing a drop-off) is a major strategic beat.

- **Source:** gold veins inside mines (the main mine near the base + expandable mines on the map).
- **Tool:** pickaxe (faster mining is a talent/meta upgrade).
- **Yield:** 1 gold per clean swing (initial value).
- **Carried:** in the backpack, capacity 10 (upgradeable).
- **Deposit:** town hall or storage building.

5.2 Wood — chopped and carried in your arms

Chop trees with the axe. A tree visibly **falls with physics** when finished, yielding logs. You gather logs into your **arms** (held in front of you) and carry them back. A full armload visibly blocks part of your view and slows you slightly.

- **Source:** trees (each yields N logs).
- **Tool:** axe (faster chopping is a talent/meta upgrade).
- **Carried:** in both arms, capacity 10 logs (upgradeable).
- **Deposit:** town hall or storage building.

5.3 Food (abstracted)

Food/upkeep exists but is **managed entirely by the RTS layer** (buildings provide it, units consume it). The player worker does **not** gather food; it only constrains the General’s economy and unit cap. This keeps the player focused on exactly two physical resources.

5.4 Gathering feel — visual references

MOCKUP

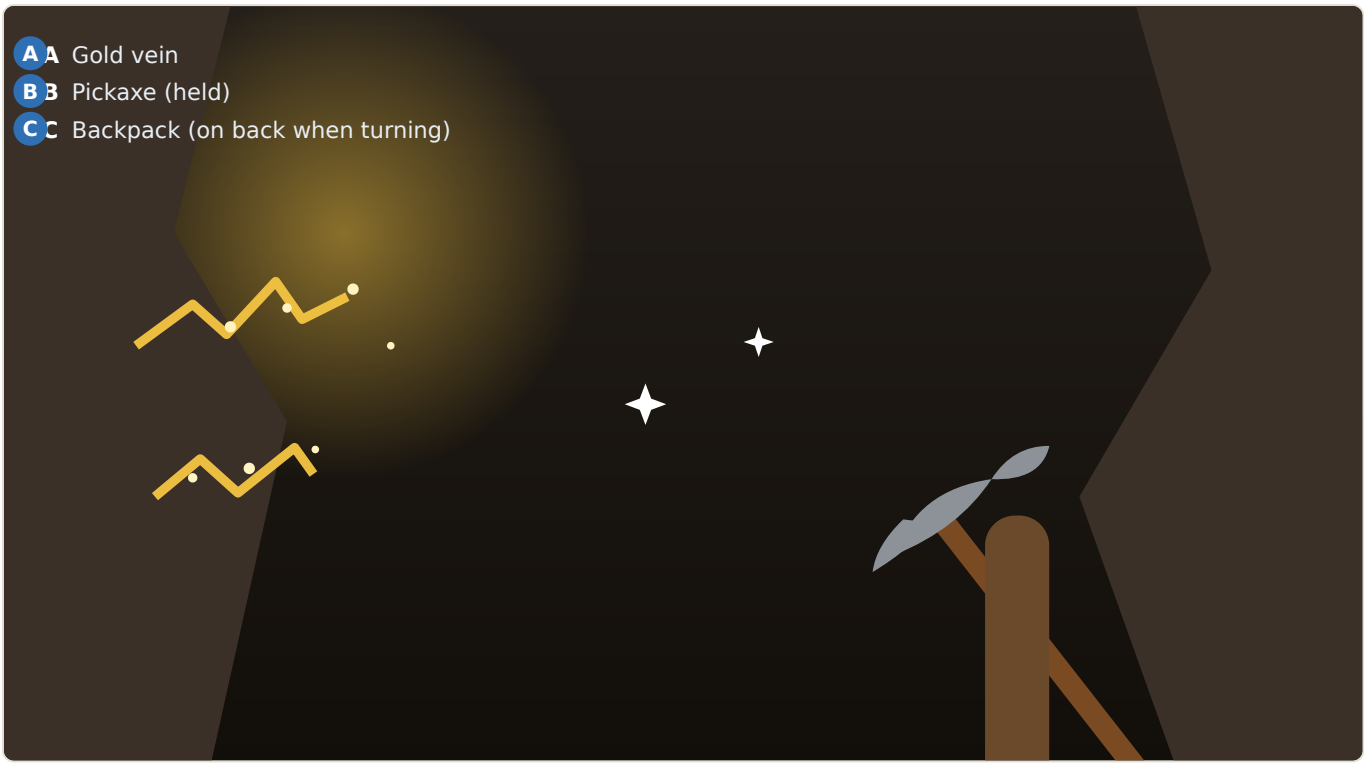


Fig. 5.1 — Inside the gold mine (first-person). Strike glowing veins (A) with the pickaxe (B) to fill the backpack (C).

MOCKUP

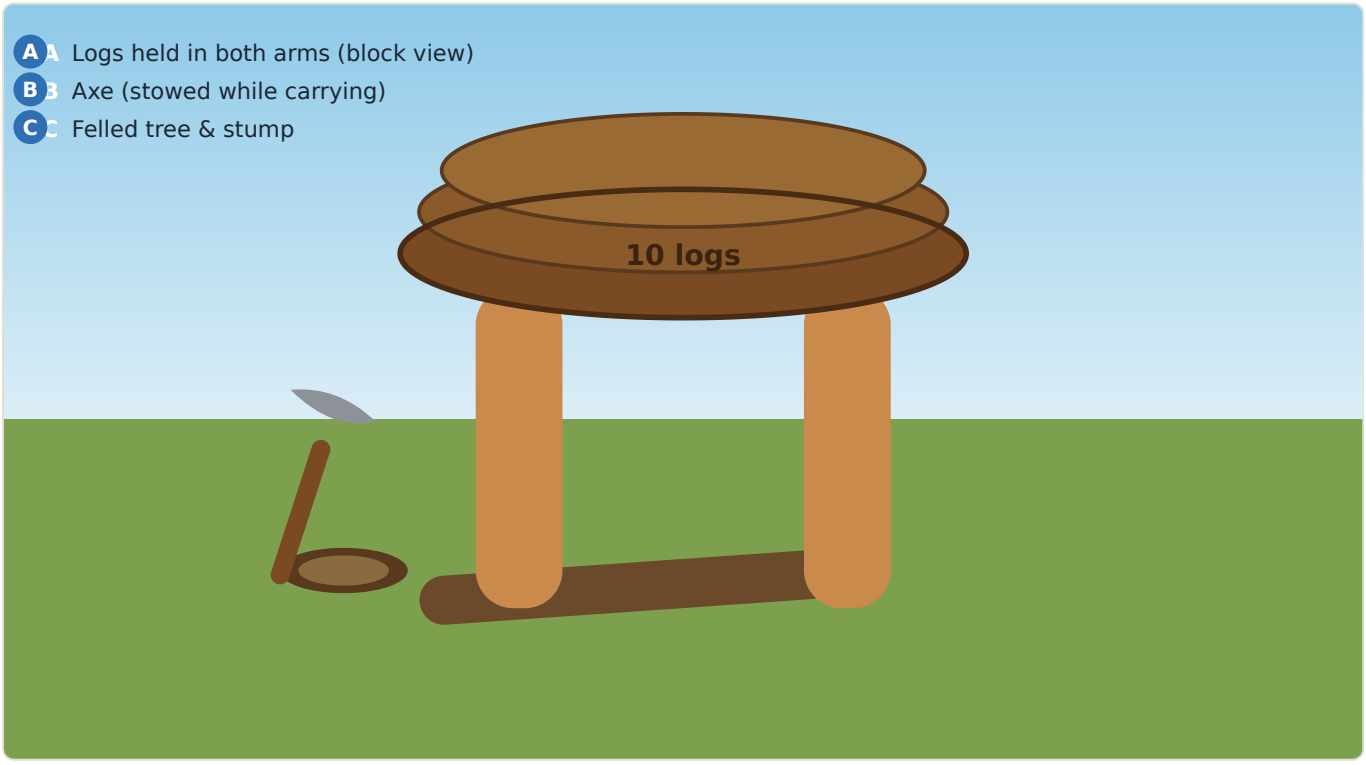


Fig. 5.2 — Carrying wood (first-person). Logs fill the view (A); the axe is stowed (B); the tree has physically fallen (C).

6 The General — RTS AI

6.1 Role & relationship

The **General** is an AI that plays the entire RTS for your faction: it builds the base, manages the economy, expands, recruits heroes, trains units, and commands the army. It is the authority figure above you. It is **not** omniscient, and it is **not** optimal — it is a fallible commander whose decisions you can observe and, sometimes, correct.

6.2 Orders & the command panel

- **Delivery:** orders appear in the **General's Orders** panel (top-left HUD), with a **sound cue** on assignment.
- **Navigation:** when the General wants you to go somewhere, it drops a **ping on the minimap** (and an in-world marker).
- **Order types:** gather gold · gather wood · build X · repair X · defend a point · attack a target · scout · colonize a mine · haul a specific stockpile.
- **Priority:** each order has a priority (primary/secondary); the panel shows at most a few active orders.
- **Acceptance is optional:** orders are never mandatory — ignoring them has loyalty consequences (§6.5), not hard control.

6.3 Intentional suboptimal behavior (the "error catalog")

By design, the General makes **detectable mistakes** that a sharp player can spot and correct via influence (§7). The error rate is scaled by difficulty. The catalog of mistakes the AI can make:

Mistake	Observable symptom	Player's possible influence
Forgets to gather a resource	Gold/wood stock sits empty or critical while workers idle.	"Prioritize gold / wood".
Fails to colonize an available mine	A rich expansion mine stays unclaimed.	"Colonize this mine" (ping it).
Leaves units idle	Trained units stand around unassigned.	"Send units here / defend / attack".
Doesn't train warriors	Enemy grows; your army stays tiny.	"Train warriors".
Skips food buildings	Hits unit cap; can't produce more.	"Build food/storage".
Doesn't recruit a hero	No hero is trained at the shrine.	"Recruit a hero".
Doesn't level the hero	Hero stays low level / unused.	"Level / heal the hero".
Doesn't heal the hero	Hero fights wounded or retreats too late.	"Heal the hero".

6.4 General's difficulty & "personality"

- Difficulty scales two things: **economic/military competence** and **error frequency** (higher difficulty = fewer mistakes = fewer easy influence opportunities).
- The General has a light **personality** (tone of messages): strict, paternal, paranoid, etc. — used mainly for flavor and to make trust/loyalty feel personal.
- `Deferred` Voice acting for the General; MVP uses text + a distinct audio cue.

6.5 Obedience / Loyalty system

Loyalty measures how much you are doing what you're told. It is a single gauge (0–100).

- **Starts** at a comfortable default (e.g., 60).

- **Gains** when you obey orders, complete tasks, and contribute to the war.
- **Drains** when you ignore orders, actively defy them, or flee a fight you were sent to.
- **No intermediate thresholds:** there is no graduated punishment. The only hard event is at **0**.

> **At zero loyalty — “you are relieved of duty”**
> When loyalty hits **0**, the General declares you a traitor and **dispatches units to kill you**. There is a short **grace window**: if you can push loyalty back above 0 before they reach you (e.g., by completing an urgent order), you are spared. Otherwise, you are hunted. Surviving an assassination attempt re-establishes a small loyalty baseline.

6.6 Confidence / Trust system

Trust measures how much the General **listens to you**. It is independent from loyalty: you can be obedient yet distrusted, or a maverick who is nonetheless respected.

- **Gains** when you accomplish **feats** during the match: killing enemies, correctly influencing him, completing hard tasks.
- **Loses** when you prove a bad soldier (repeatedly failing orders, or giving wrong advice).
- **Effect:** higher trust → the General accepts your influence (§7) more often; low trust → he rejects you or openly ignores you.

MOCKUP

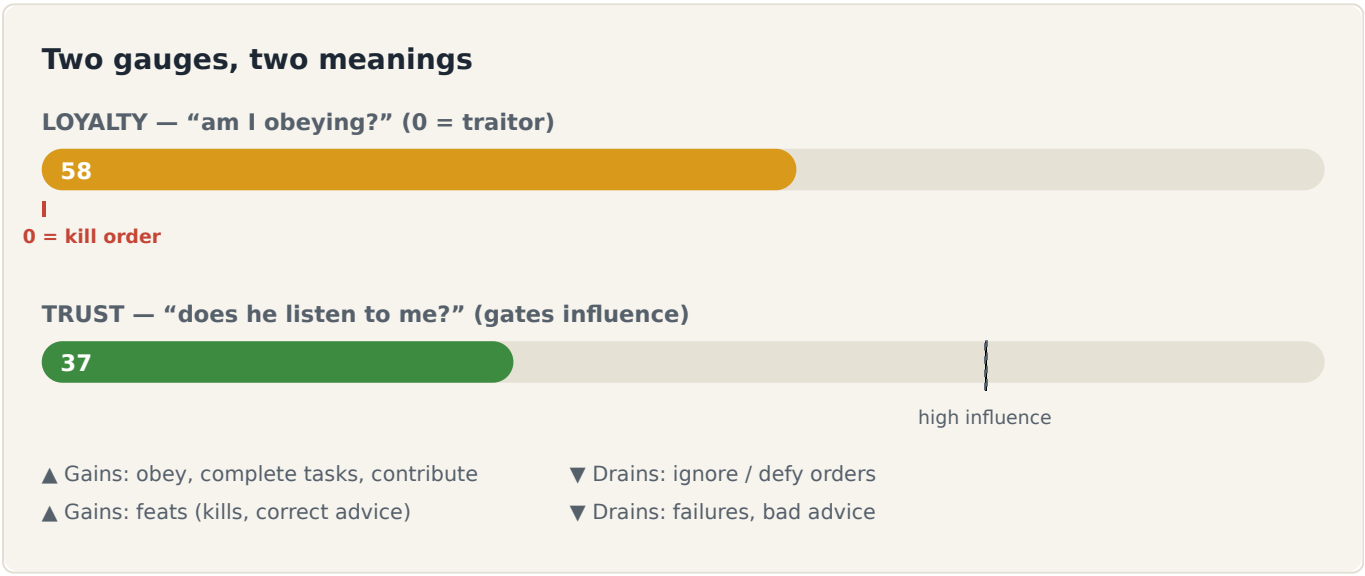


Fig. 6.1 — Loyalty (obedience) vs Trust (confidence). Loyalty gates survival; Trust gates influence. Both are shown in the HUD.

7 The Influence System

7.1 Concept

You cannot give the General orders. You can only **suggest**. Influence is indirect and persuasion-based — “like a pet nudging its owner toward the right decision.” Because the General makes mistakes (§6.3), there is always something worth nudging, and because he has a **trust** score (§6.6), your suggestions are not always accepted.

7.2 The communication wheel

Hold a key/button to open a radial **communication wheel**, choose a suggestion, then optionally aim at a location or target to attach it to. Suggestions are context-aware (e.g., “colonize mine” requires aiming at a mine).

Wheel option	Meaning / effect on the General
Attack here	Suggest attacking a location/army.
Defend here	Suggest defending a point or base.
Colonize this mine	Suggest expanding to a specific gold mine.
Build X	Suggest constructing a specific building.
Retreat	Suggest pulling back an endangered force.
Train warriors	Suggest producing more military units.
Recruit / heal hero	Suggest recruiting a hero, leveling it, or healing it.
Prioritize gold / wood	Suggest shifting economy focus to one resource.

MOCKUP

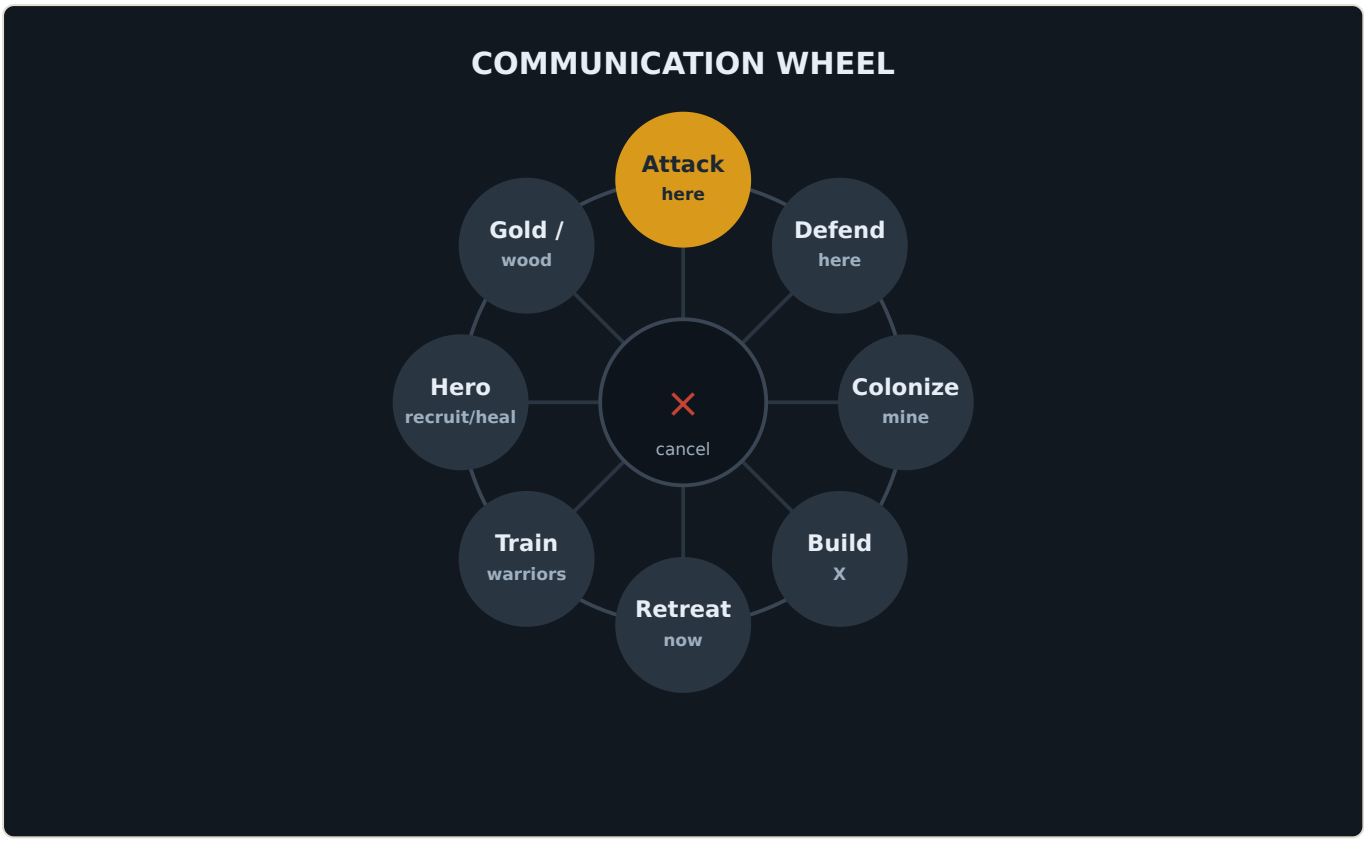


Fig. 7.1 — The communication wheel (8 suggestions + center cancel). The highlighted option (“Attack here”) shows current selection; release to send the suggestion.

7.3 Acceptance & rejection

- **Acceptance chance** is driven by the **trust** score, the **quality** of the suggestion (is it actually a good idea right now?), and a small random factor.
- On **accept**: the General updates its plan and acknowledges (“Good eye, peon.”).
- On **reject**: the General refuses (with a reason if trust is low: “You are not in a position to advise me.”).
- Suggestions that prove **correct** (they led to a better outcome) raise trust; **wrong** suggestions lower it.
- There is a short **cooldown** between suggestions to prevent spamming.

7.4 XP reward

You earn XP **when a suggestion is accepted** (and bonus XP if it later proves correct). This makes influence a real, repeatable source of progression — but gated by trust and cooldown so it can’t be farmed mindlessly.

8 Talent Trees (in-match progression)

8.1 XP & leveling

- **XP sources:** gathering resources, killing enemies, and accepted influence (§7.4).
- **Level cap:** 30. **1 talent point** per level (30 points total).
- **Spending:** open the talent screen (pause-safe) to allocate points.
- **Respec:** allowed — refunds points (with a small cost/cooldown to discourage trivial swapping).
- **On death: all in-match talents and levels reset to zero**; permanent (meta) progression is kept (§9).

8.2 The three trees

MOCKUP

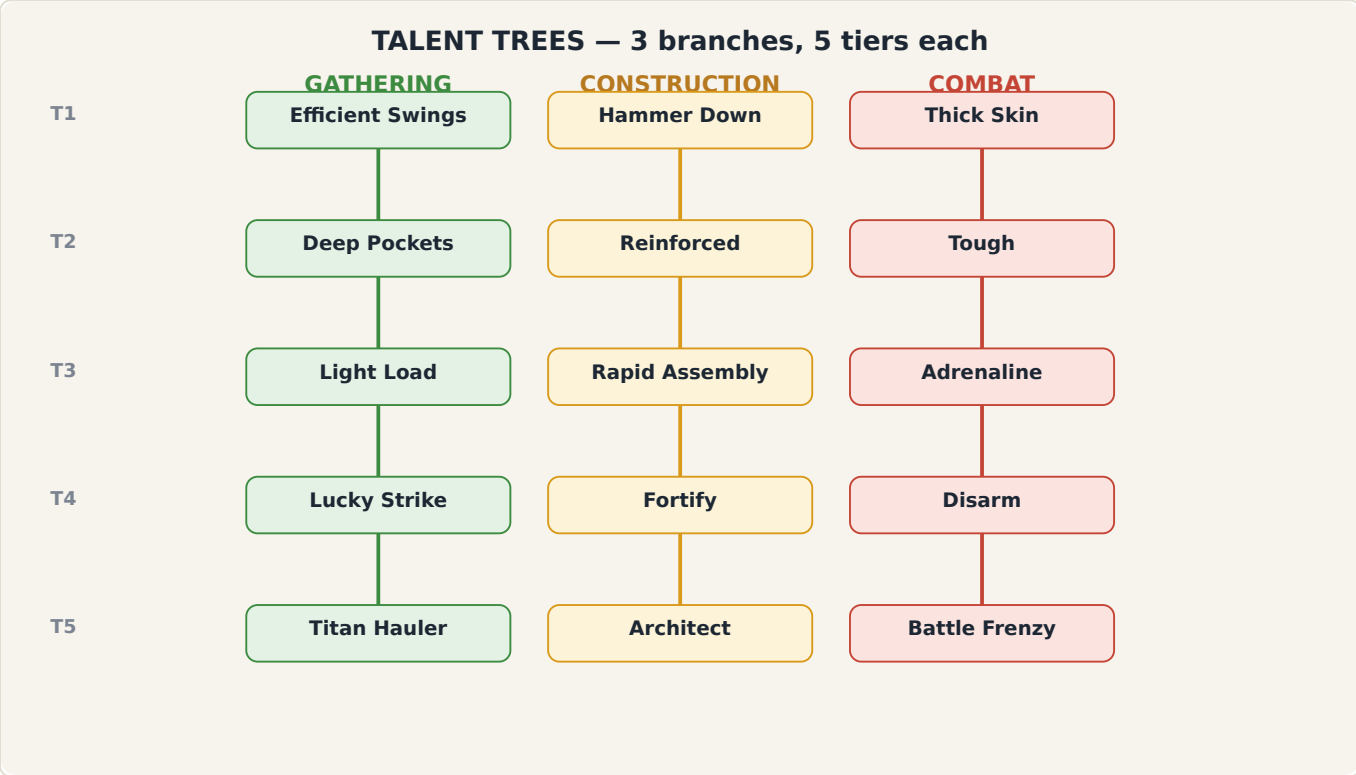


Fig. 8.1 — Talent trees (structure). Full node list below; each node can be invested in multiple times (ranks).

8.3 Full node list (initial design — extrapolated, tunable)

Tree	Tier	Node	Effect
Gathering	T1	Efficient Swings	+10% mining & chopping speed per rank.
	T2	Deep Pockets	+5 gold carry capacity per rank.
	T2	Strong Back	+5 wood carry capacity per rank.
	T3	Light Load	Reduced move-speed penalty while carrying.
	T3	Lucky Strike	Chance of bonus resource per swing.
Gathering	T4	Master Prospector	Rich veins are highlighted through walls.
	T5	Titan Hauler	Large carry-capacity boost + faster deposit.

Construction	T1	Hammer Down	+15% build speed per rank.
	T1	Careful Work	+15% repair speed per rank.
	T2	Reinforced	Buildings you construct start with bonus HP.
	T3	Rapid Assembly	+25% build speed (stacks).
	T4	Fortify	Can build temporary barricades/walls.
Construction	T5	Architect	Instantly finish one building (long cooldown).
Combat	T1	Thick Skin	+max HP per rank.
	T1	Quick Feet	+move speed per rank.
	T2	Tough	+damage reduction (armor).
	T2	Power Swing	+melee damage per rank.
	T3	Adrenaline	Faster stamina regen in combat.
Combat	T4	Disarm	Chance to wrestle an enemy’s weapon mid-fight.
	T4	Weapon Mastery	Use picked-up weapons at full effectiveness.
	T5	Battle Frenzy	Large combat boost at low HP.

Design note

The tree is intentionally **small enough to max out meaningfully at level 30** but deep enough to force choices. Exact node placement, ranks, and numbers are tuning values held in data.

9 Permanent Progression (meta)

9.1 Earning

- At the **end of each match**, you earn **Renown** (meta XP/currency) based on performance: win/loss, resources gathered, kills, influence successes, and survival.
- Renown is spent on a **permanent upgrade tree** that persists across matches, modes, deaths, and (eventually) races.
- This is the primary long-term hook of Free Play and the campaign's progression backbone.

9.2 Permanent upgrade catalog

Category	Upgrade	Effect
Tools	Faster pickaxe	Increased mining speed.
	Faster axe	Increased chopping speed.
	Heavy pickaxe	Pickaxe deals more damage (better emergency weapon).
	Sharpened axe	Axe deals more damage / cleaves.
Defense	Increased HP	+max health.
	Sturdy armor	Damage reduction (armor).
	Light armor	Armor that also improves move speed.
	Resistant armor	Armor vs magic / specific damage types.
	Regeneration	Faster out-of-combat HP regen.
Mobility	Sprint	Unlock sprint (stamina-based).
	Dodge roll	Unlock dodge roll with i-frames.
	Jump height	Higher jumps.
	Move speed	Base movement speed.
Combat	Weapon pickup	Unlock picking up weapons dropped by fallen soldiers.
	Disarm / steal	Unlock wrestling a weapon from an enemy.
	Mount theft	Unlock stealing mounts from enemies.
	Damage	Base melee/ranged damage.
Magic	Spell study	Unlock learning spells at the caster building (in-match).
	Spell power	Increased spell effectiveness.
	Spell slots	Additional equipped spells.
Utility	Carry capacity	+gold/wood carry capacity.
	Fast deposit	Deposit resources faster.
	Learning	+XP gained (faster in-match leveling).

Two layers, one identity

In-match **talents** (§8) reset on death; **permanent** upgrades (§9) never reset. A fully-upgraded permanent profile should make even a “fresh” level-1 respawn feel noticeably stronger than a brand-new player — but not so strong that death stops mattering.

10 The RTS Layer

10.1 Overview & ownership

The RTS layer is the full classic strategy game that the **General runs autonomously**. The player does **not** issue RTS commands — the player physically **does the work** (a worker is the unit that executes build/gather orders) and **influences** high-level strategy (§7). The RTS layer is what gives the world life, pressure, and context.

The structure below is **inspired by classic RTS games** (Warcraft-style: town-hall tiers, prerequisites, unit-production buildings, heroes, call-to-arms, expanding gold mines) but uses **original naming and slightly different stats**.

10.2 Buildings & tech tree (MVP — Orcs)

MOCKUP

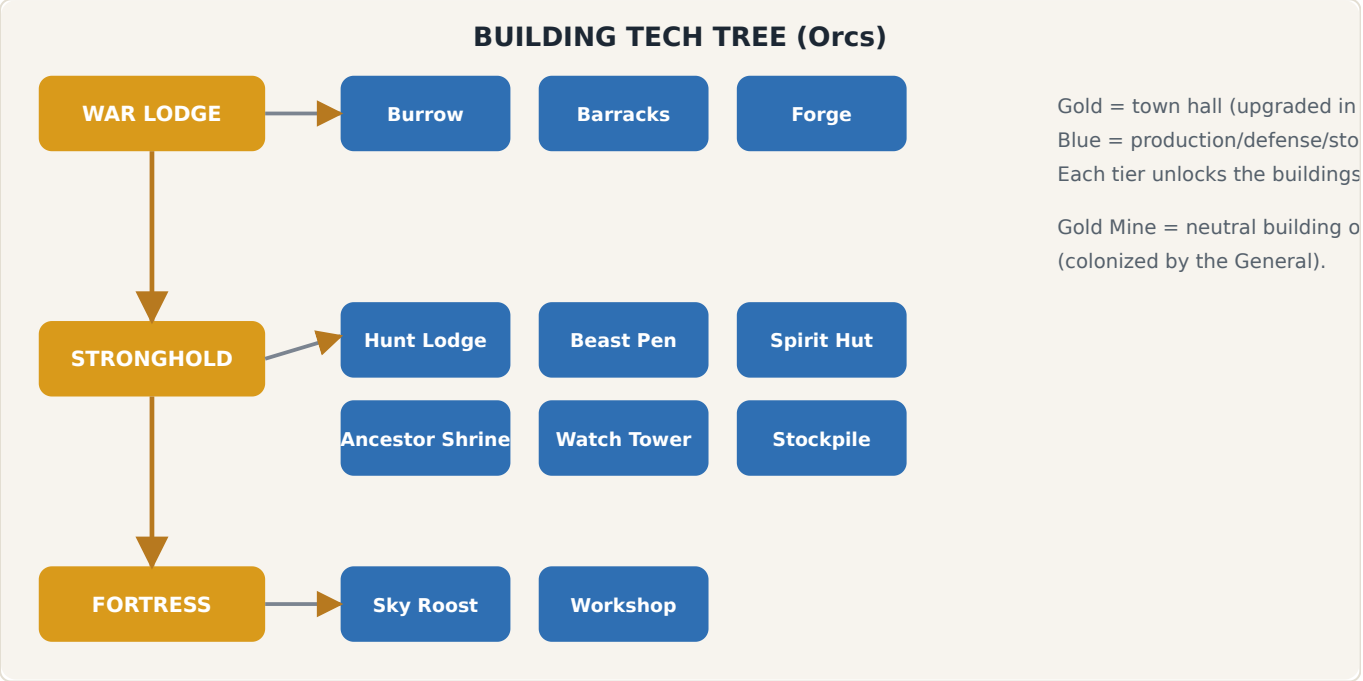


Fig. 10.1 — Building tech tree (Orcs). The town hall upgrades War Lodge → Stronghold → Fortress; each tier unlocks new buildings. Full table below.

Building	Tier	Role / output
War Lodge	1	Town hall. Trains Peons (workers), provides food, **drop-off** for gold & wood.
Burrow	1	Food + garrison (workers hide/defend inside).
Barracks	1	Trains **Bruiser** (melee infantry).
Forge	1	Attack/armor upgrades for units.
Stronghold	2	Town hall upgrade (more food, HP, unlocks Tier 2).
Hunt Lodge	2	Trains **Skirmisher** (ranged).
Beast Pen	2	Trains **Outrider** (mounted) and **Juggernaut** (heavy melee).
Spirit Hut	2	Trains casters **Stormcaller** / **Bloodspeaker**; where the player can **study magic**.
Ancestor Shrine	2	Recruits **heroes**.

Watch Tower	2	Defense (attacks nearby enemies).
Stockpile	2	Extra storage / **drop-off** + food.
Fortress	3	Final town hall upgrade (unlocks Tier 3).
Sky Roost	3	Trains flying **Sky Raider** / **Saboteur**.
Workshop	3	Trains **Demolisher** (siege).

Costs, build times, HP and food values are **inspired by classic RTS balance** but adjusted to be original; all values live in data (JSON/ScriptableObjects).

10.3 Units & roster (MVP — Orcs)

Unit	Role	Type	Produced at
Peon	Worker (gather/build/repair); the player's unit.	Worker	War Lodge
Bruiser	Front-line melee infantry.	Melee	Barracks
Skirmisher	Ranged attacker; anti-air.	Ranged	Hunt Lodge
Outrider	Fast mounted raider.	Mounted	Beast Pen
Juggernaut	Heavy melee tank.	Melee	Beast Pen
Stormcaller	Caster: offensive / support spells.	Caster	Spirit Hut
Bloodspeaker	Caster: healing / debuffs.	Caster	Spirit Hut
Sky Raider	Flying attacker / harassment.	Flying	Sky Roost
Saboteur	Flying anti-air / anti-building.	Flying	Sky Roost
Demolisher	Siege vs buildings / clumped units.	Siege	Workshop
Transport	Carries troops (ground or air).	Transport	(race-specific; deferred for MVP)

10.4 Unit types & rock-paper-scissors

MOCKUP

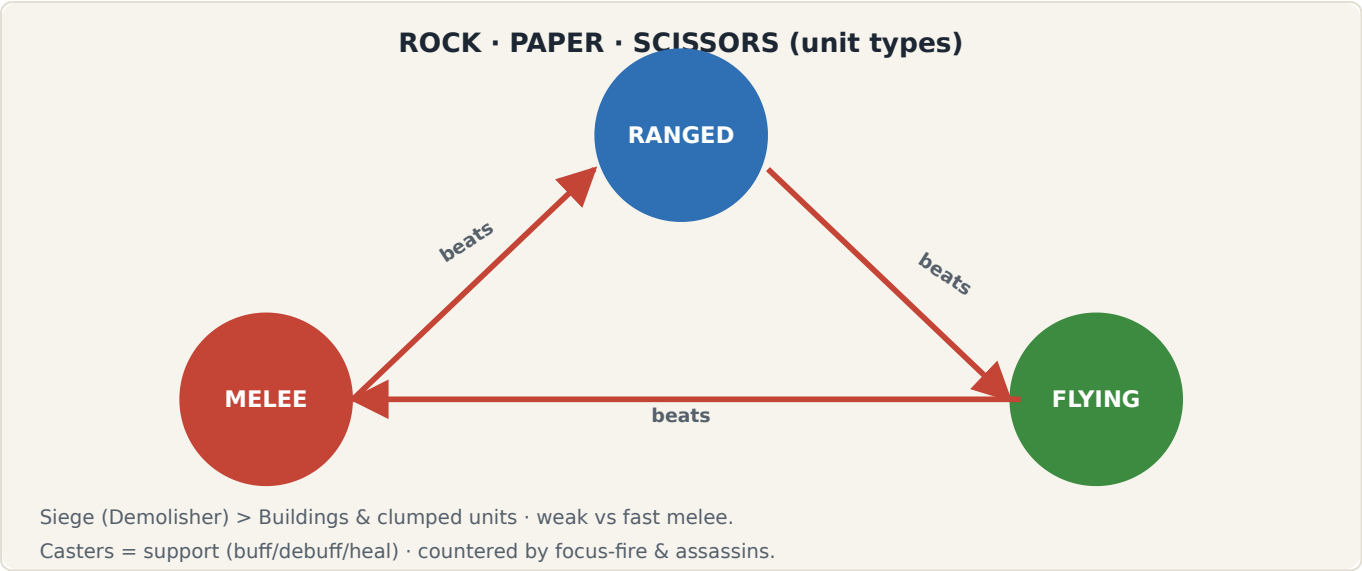


Fig. 10.2 — Core counter triangle (Melee → Ranged → Flying → Melee). Siege and casters sit outside the triangle as specialist types. Heroes are elite all-rounders countered by focus fire.

10.5 Heroes (separate from the player)

Heroes are **separate units** recruited by the General at the **Ancestor Shrine**. The player is **not** a hero — the player is a worker with its own talent tree. Heroes gain levels from combat, have abilities, and can be healed; the General can (and sometimes fails to) recruit, level, and heal them (see §6.3). MVP hero set (Orcs): **Warlord** (melee), **Storm Shaman** (caster), **Beastmaster** (summoner), **Shadowblade** (assassin).

10.6 Call to Arms (workers → militia)

The General can issue a **call to arms**: workers (including the player, if you choose to answer) temporarily become **militia** with a **boost to armor and damage** for a short duration, then revert to workers. This is the classic “desperate defense” moment, and a chance for the player to fight meaningfully.

10.7 Economy & map

- **Gold:** from gold mines (a starting mine near the base + expansion mines that must be **colonized**).
- **Wood:** from trees.
- **Food:** provided by buildings (War Lodge, Burrow, Stockpile); caps unit count.
- **Starting state:** each General begins with a few workers, a gold mine, and nearby timber.
- **Map:** sized so walking across at normal speed takes ~2 minutes; 1v1 layout.

10.8 Win condition

The match ends when an **enemy base is destroyed** (its town hall is destroyed). Other objectives (heroes, mines, armies) are means, not ends.

11 Races & Factions

11.1 Overview

Four playable races, each with a distinct worker and theme. The **MVP ships only the Orcs**; the other three are post-MVP but share the same systems, so everything must be **data-driven** to support them.

Race	Worker	Theme	Archetype (sketch)
Orcs MVP	**Peon**	Brutal, tribal, hard-hitting.	Strong melee, heavy units, straightforward rush/army focus.
Humans Deferred	**Peasant**	Versatile, disciplined, defensive.	Balanced roster, strong fortifications & siege.
Elves (Night Elves) Deferred	**Wisp**	Nature, mobility, night/fey magic.	Fast, ranged & magic-heavy, stealth.
Undead Deferred	**Acolyte**	Death, corruption, swarm.	Raises units, debuffs, attrition & magic.

11.2 The worker is the star

Regardless of race, the **worker is the player's avatar**. Each worker has the same core verbs (mine, chop, build, haul, fight) but a distinct look, tool set, and minor flavor differences. All worker abilities come from the shared talent tree (§8) and meta progression (§9).

11.3 Race balance philosophy

There is **no hard "race A beats race B" rock-paper-scissors**. Balance emerges from the **unit-type** counter triangle (§10.4) plus each race's roster strengths and weaknesses. Example: a race rich in flying units will be strong against another race that lacks anti-air ranged units — a matchup that arises naturally from composition, not from a fixed matrix.

> **Architecture requirement**
>
> Because three races are deferred, the entire RTS layer (buildings, units, costs, upgrades) and the worker configuration (tools, looks, stats) must be **data-driven and race-agnostic**. Adding a race should be content work, not engine work.

12 Physics & Game Feel

12.1 Why it matters

“Superb physics” and “satisfying gameplay” are explicit pillars (§1.4, pillar 4). The feel of chopping, mining, carrying, and fighting is what sells the fantasy. This section defines concrete feel requirements.

12.2 Harvesting physics & feedback

Interaction	Feel requirements
Chopping trees	Visible wood chips; the tree sways with each hit; on the final hit it **falls with physics** (ragdoll-style rigidbody), lands, and yields logs. Strong haptic/sound feedback.
Mining	Pickaxe swing animation; sparks and dust on impact; gold glints fly off; the vein visibly depletes.
Carrying	Logs visibly fill the arms (weighty, blocking view); backpack visibly bulges with gold; movement subtly heavier with a full load.
Building	A scaffold appears and visibly assembles as you work; hammer hits produce sparks; progress is physical, not just a bar.

12.3 Combat feel

- Melee swings have **weight** (wind-up, impact, recovery) and hit-stop on connect.
- Ragdolls on death (units crumple with physics, not scripted falls).
- Dodge roll and sprint feel responsive; body awareness keeps the player grounded.

12.4 Technical approach

- Use Unity Physics for characters, ragdolls, and the falling-tree rigidbodies.
- Use animation events + particle systems for impact feedback (chips, sparks, glints).
- Tune “game feel” (hit-stop, camera shake, screen effects) via a single **FeelManager** so values are centralized.
- `Deferred` Advanced cloth/fluid physics; MVP uses hand-authored effects.

13 Art & Audio Direction

13.1 Art style

Stylized cartoon fantasy in the spirit of World of Warcraft: exaggerated proportions, chunky readable silhouettes, bold saturated colors, soft hand-painted textures, and clear readability at a distance. Serious stakes, playful presentation. See §21 for reference media.

- **Characters:** oversized hands/weapons, strong silhouettes, readable faction colors.
- **Environment:** painterly terrain, stylized trees/rocks, glowing gold veins for gameplay readability.
- **UI:** clean, diegetic-warm HUD (see Fig. 4.1); clear iconography for orders, resources, and gauges.
- **Palette:** faction colors (Orc red/earth, Human blue/gold, Elf green/silver, Undead purple/bone) + gold and wood resource colors (yellow/gold, brown).

13.2 Audio

- **SFX:** satisfying harvest/impact sounds (tree fall, pickaxe clink, deposit), combat (swings, hits), UI cues (order received = distinct chime).
- **Music:** adaptive score that shifts with tension (calm base-building → battle). Fantasy orchestral/percussive tone.
- **Voice:** Deferred — MVP uses text + character-distinct audio cues for the General.

13.3 Audio & art asset sourcing

Assets will be **purchased** (stylized/cartoon packs) and adapted, not produced in-house for the MVP. The asset list (§19) enumerates what is needed.

14 Controls & Input

14.1 Input map

Action	PC (keyboard/mouse)	Gamepad (SteamDeck)	Touch (Android)
Move	WASD	Left stick	Virtual stick
Look / aim	Mouse	Right stick	Touch-drag
Jump	Space	A (South)	Button
Sprint	Shift	L3 / hold stick	Button
Dodge roll	Ctrl / double-tap	B (East)	Button
Interact / gather / build	Left click / E	RT / X (West)	Tap action button
Attack	Left click	RT	Attack button
Communication wheel	Hold Q	Hold LB	Hold wheel button
Talent screen / menu	Tab / Esc	Start / Select	Menu button

14.2 Requirements

- All controls must be **remappable** (MVP: PC; post-MVP: gamepad/touch).
- Use Unity’s **Input System** package with an abstraction layer so the same actions drive keyboard, gamepad, and touch.
- Touch controls are `Deferred` (Android), but the input architecture must not block them.

15 Multiplayer & Networking

15.1 Model

- **Mode:** PvP **1v1** (two camps).
- **Players per camp:** up to **3 human workers** (6 max total).
- **No human General:** each camp's General is always an AI. Humans are always workers.
- **Cooperation:** workers of the same camp share the objective and can coordinate (via the communication wheel toward the General and toward each other).

15.2 Network architecture

- **Topology: Peer-to-peer (P2P)**, no dedicated server for the MVP of multiplayer.
- **Authority:** one peer is the **host** and owns the authoritative simulation (RTS layer + both Generals' AI + unit combat). Worker inputs are sent to the host; the host broadcasts state.
- **Framework:** Unity **Netcode for GameObjects** (or an equivalent P2P-friendly solution).
- **Determinism:** the General AI should be **deterministic** given the same seed so it can be re-simulated/verified.

> **Deferred for MVP**
>
> Multiplayer (netcode, P2P, host authority, sync) is a **large independent effort** and is **deferred** to post-MVP. The architecture must still be written to **not preclude** it: keep the RTS simulation and the worker controller separable from the local-player input path.

16 Technical Specification

16.1 Engine & packages

Item	Choice
Engine	Unity 2022 LTS (or newer), **Universal Render Pipeline (URP)**.
Language	C#
Input	Unity **Input System**.
Networking (later)	Unity Netcode for GameObjects.
Physics	Unity Physics (built-in / PhysX) + ragdolls.
Content pipeline	ScriptableObjects + JSON for all tunable data.
Save	Local JSON save for meta progression & settings.

16.2 Performance targets

Platform	Target
Windows (MVP)	60 FPS on a mid-range PC at 1080p, with a full RTS battle + physics in progress.
Steam Deck (post-MVP)	40–60 FPS, 800p, Low/Medium settings.
Android (post-MVP)	30–60 FPS on mid-range devices, Low settings, reduced draw distance/particles.

Performance discipline: the RTS simulation (units/buildings) must be **cheap enough to run alongside a full 3D first-person scene**. Use LODs, object pooling, and avoid per-frame allocations in the simulation.

16.3 High-level architecture

System	Responsibility
RTS Simulation	Economy, buildings, units, tech tree, combat resolution. Runs independently of the player.
General AI	Planner that issues orders, manages the base, makes (fallible) decisions, accepts/rejects influence.
Worker Controller	First-person player: locomotion, body awareness, carrying, tools, combat.
Influence System	Communication wheel, suggestion evaluation, trust/loyalty gauges.
Progression	XP/levels, talent trees (in-match), Renown/meta upgrades (persistent).
Feel System	Camera shake, hit-stop, particles, haptics (centralized tuning).
UI	HUD, orders panel, minimap, communication wheel, talent/meta screens.

16.4 Data-driven content

- All tunable values (unit stats, building costs, talent effects, meta upgrade values, General AI parameters) live in **data assets**, not code.
- This is mandatory to (a) tune balance quickly and (b) add the 3 deferred races as pure content.

16.5 Save & persistence

- Persist: meta progression (Renown + purchased upgrades), settings, campaign progress.
- Do **not** persist in-match state (talents/levels reset by design).

17 MVP Scope

17.1 In scope (MVP — “the vertical slice”)

Area	Deliverable
Platform	Windows build (keyboard/mouse + gamepad).
Race	Orcs — the **Peon** worker only.
Modes	**Free Play** (1–2 maps, difficulty tiers) + a **minimal Campaign** (a short 3–5 mission arc).
Worker	First-person locomotion, body awareness, sprint, jump, gather (gold in mine + wood), carry (backpack / arms), deposit, build/repair, basic combat (axe/pickaxe).
General AI	Orders panel + sound cue, minimap pings, a working **subset** of the error catalog, loyalty gauge (kill at 0), trust gauge.
Influence	Communication wheel (all 8 options), accept/reject logic gated by trust, XP on accept.
Talent trees	3 trees (Gathering/Construction/Combat), level cap 30, respec, reset on death.
Meta progression	Renown earned at match end + a **subset** of permanent upgrades (tools, HP, speed, sprint/roll unlocks, weapon pickup).
RTS layer	Orc buildings/units/tech tree, heroes (recruit/level/heal), call-to-arms, expanding gold mines, win = destroy enemy base.
Feel	Physics falling trees, mining feedback, visible carrying, ragdoll deaths.
UI	Full HUD (Fig. 4.1), communication wheel, talent & meta screens.
Persistence	Local save for meta progression, settings, campaign progress.
Audio	Core SFX + music (stock/purchased assets acceptable).

17.2 Out of scope for MVP (deferred)

Area	Why deferred
Multiplayer PvP	Netcode/P2P is a large independent effort (§15).
Other 3 races	Content work; architecture is data-driven to support them.
Android & Steam Deck ports	Perf + touch UI + certification work.
Touch controls	Tied to Android.
Voice acting / cinematics	MVP uses text + simple camera staging.
Full campaign (many missions)	MVP ships a short arc to prove the format.
Transport units	Race-specific content; later.

17.3 Explicitly cut (out of scope entirely for now)

- Climbing / vertical traversal.
- Third-person camera mode.
- Level/map editor and modding tools.

- Spectator / replay system.

18 Roadmap (post-MVP)

Phase	Focus
Phase 1 — MVP	Vertical slice: Orcs, Free Play + minimal Campaign, full worker loop, General AI, influence, talent + meta, RTS layer, physics feel. Windows only.
Phase 2 — Multiplayer	PvP 1v1 (up to 3 workers/camp), P2P + host authority, netcode, basic lobby.
Phase 3 — Races	Humans, Elves, Undead (workers + rosters + buildings), race balance pass.
Phase 4 — Steam Deck	Gamepad polish, verified performance, Steam Deck certification.
Phase 5 — Android	Touch controls, mobile optimization, store release.
Phase 6 — Content & polish	Full campaign, more maps/modes, voice acting, meta expansion, live balance.

19 Asset & Content List

Assets are purchased (stylized/cartoon packs) and adapted. This is the minimum set for the MVP.

Category	Items
Characters	Peon (worker) with first-person hands/arms; Orc units (Bruiser, Skirmisher, Outrider, Juggernaut, Stormcaller, Bloodspeaker, Sky Raider, Saboteur, Demolisher); 4 heroes (Warlord, Storm Shaman, Beastmaster, Shadowblade); enemy faction stand-ins.
Buildings	War Lodge/Stronghold/Fortress, Burrow, Barracks, Forge, Hunt Lodge, Beast Pen, Spirit Hut, Ancestor Shrine, Watch Tower, Stockpile, Sky Roost, Workshop, Gold Mine.
Environment	Terrain, trees (with falling state), rocks, gold veins, mine interior, props.
Tools / weapons	Axe, pickaxe, hammer, plus soldier weapons (melee + ranged) for pickup.
Animations	Worker: walk/run/sprint/jump/roll, chop, mine, carry (wood/gold), build/repair, melee swing, pickup, death/ragdoll. Units: combat/idle/move/death. First-person arms for all worker actions.
VFX	Wood chips, mining sparks/dust, gold glints, tree-fall dust, hit impacts, spell effects (heroes/casters), building-construction scaffold.
Audio	SFX: tree fall, pickaxe/axe/hammer impacts, deposit, UI chime (order), combat hits, death. Music: base-building and combat tracks.
UI	HUD elements, order icons, minimap assets, communication-wheel icons, talent/meta screen layout, fonts.

20 Acceptance Criteria

Each feature is “done” only when its acceptance criteria pass. These are the MVP gates.

20.1 Worker & gathering

- Player can walk/run/sprint/jump and sees hands, arms, feet and body in first person.
- Chopping a tree plays feedback and the tree **physically falls**, yielding logs the player can pick up.
- Mining inside the mine yields gold into the backpack; capacity is enforced and visibly shown.
- Carried wood blocks part of the view; carrying slows the player; deposit banks resources.

20.2 The General & influence

- Orders appear in the panel with a sound cue; minimap pings point to destinations.
- Ignoring orders drains loyalty; at 0 the General sends units to kill the player (with a grace window).
- The communication wheel sends suggestions; acceptance is trust-gated; accepted suggestions award XP.
- The General demonstrably makes at least 3 distinct mistakes from the error catalog that the player can correct.

20.3 Progression

- XP is earned from gathering, kills, and accepted influence; leveling grants a talent point; level cap 30.
- All three talent trees are spendable; death resets in-match talents/levels but keeps permanent upgrades.
- Renown is earned at match end and spent on permanent upgrades that persist across sessions.

20.4 RTS layer & match

- The General autonomously builds a base, trains units, expands to mines, recruits/levels/heals heroes.
- The match ends when an enemy base is destroyed; win/loss is reported and Renown awarded.

20.5 Technical gates

- Windows build runs at 60 FPS (1080p, mid-range) with a full battle in progress.
- All tunable values are in data assets (no hard-coded balance numbers).
- Meta progression and settings survive an application restart.
- No crash / soft-lock when the player dies and respawns mid-match.

21 References & Media

This section gives the development AI (and any artist/designer) concrete visual anchors. Screenshots below are from **open-source games** and illustrate **mechanics, camera, and UI** — not the target art style. The **art style target is World-of-Warcraft-style cartoon fantasy**, referenced by links in §21.3 (actual WoW/Warcraft imagery is copyrighted, so it is linked, not embedded).

21.1 The RTS layer (what the General “plays”)

REFERENCE



Fig. 21.1 — 0 A.D. (open-source RTS): top-down base-building, units, and economy. This is the **conceptual** view of the RTS layer the General runs, which the player sees from the inside. © Wildfire Games, CC BY-SA 3.0.

REFERENCE



Fig. 21.2 — 0 A.D.: the scale of an army clash. In Pawn, the player stands **inside** a battle like this, in first person. © Wildfire Games, CC BY-SA 3.0.

21.2 First-person gathering, mining & combat

REFERENCE



Fig. 21.3 — Minetest (open-source): first-person mining/gathering with visible tools and blocks. Reference for the **feel of mining and gathering in first person**. © Minetest contributors, CC BY-SA 3.0.

REFERENCE



Fig. 21.4 — Xonotic (open-source FPS): first-person combat movement and weapon view. Reference for **first-person combat feel and body awareness**. © Xonotic, GPL.

REFERENCE



Fig. 21.5 — Veloren (open-source): stylized low-poly fantasy landscape. Reference for a **stylized (non-photoreal) fantasy world** — a step toward the cartoon target. CC0.

21.3 Art style target (linked — copyrighted)

- **World of Warcraft** (Blizzard) — the cartoon-fantasy art style target: worldofwarcraft.blizzard.com and its media/art galleries.
- **Warcraft III: Reforged** (Blizzard) — the classic RTS structure (town hall tiers, workers, heroes, gold/wood economy) that the RTS layer is inspired by: warcraft3.blizzard.com.

21.4 Video references (watch for look & feel)

- **Warcraft III / Reforged** gameplay — RTS loop, workers, heroes, gold/wood: official Blizzard trailers on the link above.
- **Deep Rock Galactic** — first-person mining feel, entering a mine, hauling ore.
- **Medieval Dynasty** — first-person chopping, carrying logs, building.
- **Chivalry 2** — first-person melee combat weight and body awareness.
- **Valheim** — first-person tree chopping and the satisfying *tree-fall* moment.
- **Satisfactory** — first-person resource gathering, mining, and carry/automation loop.

These are recommendations; the developer should source official trailers/gameplay clips from the publishers' channels. The embedded screenshots above are all from open-source projects under permissive/free licenses, used here for internal reference and attributed.